Nafeez Ahmed Noor Ahamed

Aeronautical Engineer · Laser Gas Sensing · Sustainable Aviation

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Zwickau, Germany | German residence permit · EU Blue Card eligible · Available from 09/2027

PROFESSIONAL SUMMARY

Aeronautical engineer (B.Tech.) completing an M.Eng. at WHZ Zwickau (09/2027), based in Germany since 2023. At **Fraunhofer IWS**, develops **laser-based gas leak detection (TDLAS/WMS)** with **Python signal processing** and compensated measurement. Also experienced in **LCA**, **SAF policy** and **CSR/ESG reporting**. Thesis on **hydrogen safety in aviation**.

KEY HIGHLIGHTS

- **Laser gas sensing:** TDLAS/WMS research at Fraunhofer IWS in a BMBF-funded leak-detection project
- **Measured results:** found and fixed a filter instability; ran a 5-run statistical benchmark of acquisition hardware
- **Sustainability:** LCA/GIS air-freight study with SAF proposals, adopted as a case study; CSR/ESG reporting
- **Aviation & leadership:** aircraft and drone design, drone pilot license, four years as class representative

WORK EXPERIENCE

Research Assistant – Optical Gas Sensing (part-time), Fraunhofer Institute for Material and Beam Technology IWS
07/2026 – Present | Zwickau, Germany

- Develop laser-based gas leak detection using **Tunable Diode Laser Absorption Spectroscopy (TDLAS)** and **Wavelength Modulation Spectroscopy (WMS)** in **CLeo**, a **BMBF-funded** project on AI-based optical leak detection.
- Developed Python packages for **methane (CH4) concentration retrieval (WMS 2f/1f)** on Red Pitaya with **BMP280 temperature/pressure compensation**; found and fixed an inverted ideal-gas density correction, verified results against reference conditions and extended the calibration to other gases.
- Benchmarked **six digital filters** at 15.625 MS/s: standard IIR coefficients diverged by more than **90 orders of magnitude** at the board's **1,562x oversampling ratio**, so all filters moved to stable **second-order sections**; every Python filter needed **4–50x the acquisition time**.
- Compared Red Pitaya **Gen1 and Gen2** over **5 repeated runs** (mean ± std): Gen2 ran **3.4x faster** with a nearly identical **noise floor (about -120 dBV)** and no repeatability gain; traced invalid **SNR/ENOB** values and **shared-ADC crosstalk**.
- Characterised **photodetector bandwidth (-3 dB)**, designed a **3D-printed sensor housing** (Fusion 360), applied **Random Forest** leak classification and wrote a **43-page technical manual**.

Research Assistant – Atmospheric Plasma Technologies (part-time), Westsächsische Hochschule Zwickau
06/2025 – 10/2025 | Zwickau, Germany

- Ran controlled experiments on **50+ samples** in **cold-plasma surface treatment** research for microbial decontamination, supervised by **Prof. Dr. Fröhlich** (formerly **DLR** Institute for Materials Research).
- Optimised experimental parameters, **reducing cycle time by 25%**; coordinated lab equipment and material orders.

Production Assistant (part-time student job), Intersnack Deutschland SE
01/2024 – 09/2024 | Alsbach-Hähnlein, Germany

- **Quality and quantity checks** on a food packaging line, machine operation and **safety protocols** in a German production team.

Consultant Intern – Civil Engineering & CSR/ESG, mBYOM Consultant and Management Services LLP
09/2023 – 11/2023 | Chennai, India

- Analysed **105 technical documents** for **compliance gaps** and wrote **CSR/ESG research reports** for **management decisions**.
- Supported planning and **building-code inspections** for educational building projects with engineers and architects.

PROJECTS

Sustainable Air Freight: Water–Food–Energy Nexus | 12/2025 – 02/2026

- **LCA/GIS** study of a perishable air-freight supply chain: identified spoilage hotspots above 6% and proposed **SAF incentive policies**; adopted as a WHZ **case study**.

Supersonic Aircraft Conceptual Design (B.Tech. design project) | 01/2022 – 12/2022

- **Conceptual design** of a single-seat supersonic turbojet aircraft (mid wing, bicycle landing gear, twin tail); structural and aerodynamic analysis in **SolidWorks** and **ANSYS**.

Perching Landing Gear for a Drone (B.Tech. final project) | 01/2023 – 05/2023

- Translated bird-leg anatomy into a **cable-driven, underactuated grasping mechanism** in sustainable thermoplastics; **CAD** models, **ANSYS** analysis and a **3D-printed** prototype tested on different surfaces.

EDUCATION

M.Eng. Advanced Green Engineering and Sustainable Management, Westsächsische Hochschule Zwickau
10/2024 – Expected 09/2027 | Zwickau, Germany

- **Master's thesis (from 03/2027):** multi-sensor gas leak detection with environmental compensation in a **Herriott-cell TDLAS/WMS** setup (Red Pitaya, Teensy 4.0, pressure/temperature sensors); methane as surrogate gas, extrapolated to **hydrogen** detection requirements for **hydrogen safety in aviation**.
- Current grade **2.0**; modules include **low-carbon energy systems**, **vacuum and plasma engineering** and **life-cycle assessment**.

M.Sc. Mechanics (two semesters, 20 ECTS; moved to WHZ), Technische Universität Darmstadt
11/2023 – 09/2024 | Darmstadt, Germany

- Fracture mechanics, computational plasticity and interfacial flow dynamics; numerical modelling in **Python** and **R**.

B.Tech. Aeronautical Engineering, B.S. Abdur Rahman Crescent Institute of Science and Technology
08/2019 – 03/2023 | Chennai, India

- **CGPA 8.41/10 (German grade 1.7)**; coursework in **aerodynamics**, **flight dynamics**, **avionics** and air traffic control regulations.

LEADERSHIP & VOLUNTEERING

- **Class Representative**, B.Tech. Aeronautical Engineering (2019–2023): represented the batch for four years.
- **Student Coordinator, DARE '22** (2022): organised a health awareness event and secured high-profile guest speakers.
- **Volunteer, SAEISS drone workshop and drone seminar** (2022): event support and hands-on student drone assembly.

SKILLS

Laser spectroscopy & instrumentation: TDLAS, WMS (2f/1f), gas sensing, leak detection, calibration, Red Pitaya (SCPI), I2C sensors, FPGA

Signal processing & data: Python, MATLAB, SQL, digital signal processing (DSP), machine learning (Random Forest)

Design & simulation: Fusion 360, SolidWorks, CATIA, ANSYS, 3D printing

Sustainability & aviation: life-cycle assessment (LCA), GIS, Sustainable Aviation Fuel (SAF), CSR/ESG reporting, urban air mobility

LANGUAGES & INTERESTS

Languages: English – C1 (IELTS) · German – A2 (telc B1 in preparation) · Urdu – native · Tamil – native

Interests: football (WHZ campus league) · basketball · poetry (published in three anthologies, 2019–2020) · drones

CERTIFICATIONS & LICENSES

Urban Air Mobility (Technical University of Munich)

Hack Your Carbon Impact: Build and Publish an Emissions Tracker with MATLAB (MathWorks)

Remote Pilot License – small-category drones/UAS (DGCA India, valid 11/2022 – 11/2032)